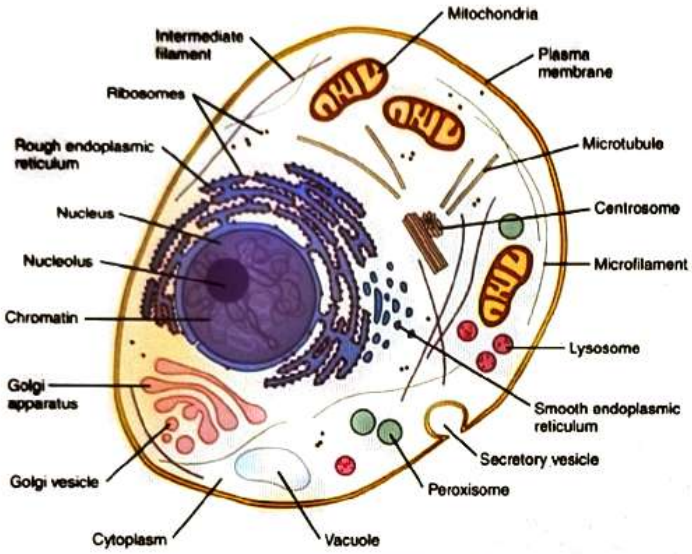
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CELLULAR PHYSIOLOGY

----- Active space --

Cell is the fundamental unit of all living organisms.



Eukaryotic Human Cell

Cell Membrane

00:01:13

AKA plasma membrane.

FUNCTION :

It is a outer barrier with selective permeability in cell.

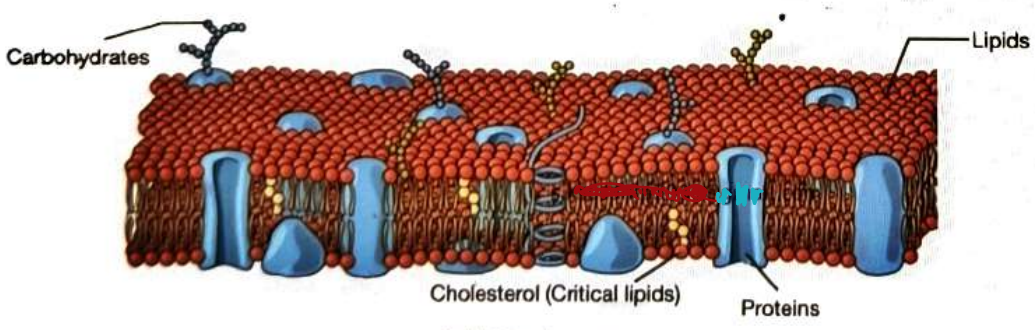
STRUCTURE :

Fluid mosaic model of cell membrane. Proposed by Singer and Nicolson.

COMPOSITION :

Macromolecules.

1. Proteins (55%) : Arranged in mosaic pattern in cell membrane.
2. Lipids (40%) : Responsible for fluidity of cell membrane.
3. Carbohydrates (5%).



Cell membrane

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Lipids

00:04:40

INTRODUCTION :

Structure :

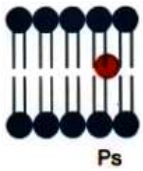
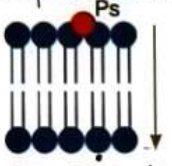
- Bilayer : Outer and Inner lipids.
- Asymmetry : Arranged specifically either on inner or outer side.

Types :

- Phospholipids (PO_4^{3-} group).
- Glycolipids (Carbohydrate side chain).
- Cholesterol (Critical lipid).

PHOSPHOLIPIDS :

Major lipid.

Types	Site	Significance
Phosphatidyl choline (Pc) AKA lecithin	Lungs : Surfactant	• Prevent alveolar collapse. • Major lipid in surfactant.
Sphingomyelin	• Nerve cell membrane. • Surfactant.	Minor lipid in surfactant
Phosphatidylinositol (IP3)	Endocrine cells	• Part of second messenger system (IP3 → DAG → Ca^{2+})
Phosphatidylserine (Ps)	Inner surface of cell membrane in all cells 	If expressed on outer surface.  Eat me Signal • Apoptosis (Programmed cell death). • Detected by Annexin V : Binds & forms complex with Ps. (Annexin V staining).
Cardiolipin (CL)	Heart : Mitochondria	In syphilis : Serum anti-CL antibody + (Detected by VDRL test)

Note :

L (Lecithin) / S (Sphingomyelin) ratio :

- To assess fetal lung maturity.
- Mature lung : ≥ 2.

GLYCOLIPIDS :

Types	Site	Significance
Cerebrosides	CNS	-
Gangliosides	GIT	GM-1 gangliosides : Receptors for cholera toxin-Fragment B.

Feedback

CHOLESTEROL :

----- Active space -----

Site : Cell membrane of skin.

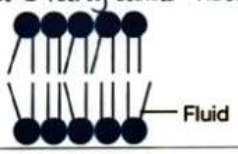
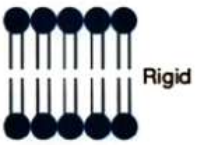
Significance :

- Precursor of Vit.D.
- 7-dehydrocholesterol $\xrightarrow[(Skin)]{Sunlight}$ Vit.D
- **Fluidity buffer** : Maintains membrane fluidity in optimal range.

MEMBRANE FLUIDITY :

Lubricative property of lipid → Maintain fluidity of membrane → Allows mobility.

Characteristics of Fluidity :

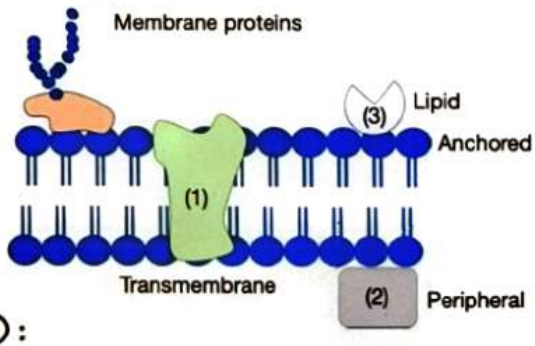
	Increased Fluidity	Decreased Fluidity
Effect on health	Good	Bad
Type of fatty acids	Unsaturated (Essential)	Saturated (Trans)
Examples	• Linoleic acid • Linolenic acid • Arachidonic acid • Omega-3 fatty acids : Abundant in fish 	• Stearic acid • Palmitic acid : (Abundant in junk food) 

Proteins

00:24:00

Classification based on location in cell membrane :

1. Transmembrane proteins/integral membrane proteins : Present throughout.
2. Peripheral proteins : Present peripherally.
3. Lipid-anchored proteins : Requires lipids for attachment.



TRANSMEMBRANE PROTEINS (INTEGRAL MEMBRANE PROTEIN) :

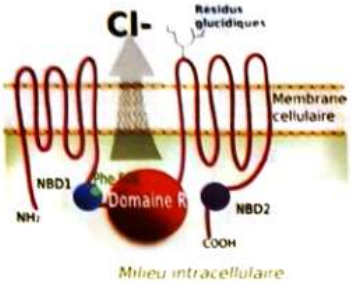
Most important.

Functions	Examples
Hormone receptors (m/c)	GPCR (G-protein coupled receptors).
Pump channels	Na^+-K^+ ATPase
Ion channels	• Cl^- channel (CFTR). (Cystic fibrosis transmembrane conductance regulator.) • In cystic fibrosis → CFTR mutation (+).

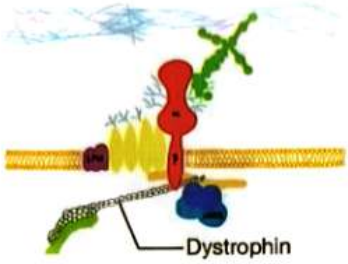
Feedback

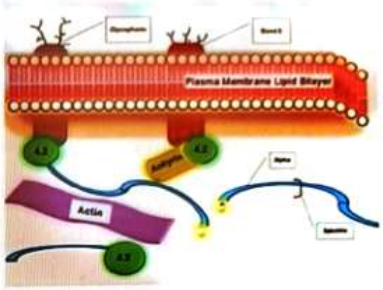
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----- Active space -----



Transmembrane protein :
Cl^- channel (CFTR)





Peripheral protein

PERIPHERAL PROTEINS :

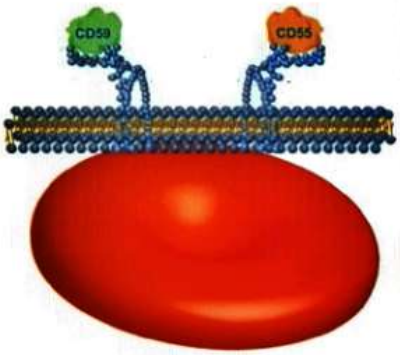
Functions :

1. Support shape of cell :

Cell	Protein	Diseases due to mutations
RBC (Biconcave disc)	Spectrin	Elliptocytosis
	Ankyrin	Spherocytosis
Skeletal muscle	Dystrophin	• Duchenne muscular Dystrophy (DMD). • Muscular weakness. Typical presentation. • Gower's sign : Climb on own body to stand up (Knee → Hip → Stand). • Respiratory paralysis (weakness of diaphragm) → Death.

2. Cell surface receptors.

LIPID ANCHORED PROTEINS :



Lipid anchored proteins in RBC.

In Rbc :

- PIGA (Phosphatidylinositol Glycan-A) gene synthesises proteins :
  - CD55 (Decay accelerating factor).
  - CD59 (membrane inhibitor of reactive lysis).
  - C8 binding protein.
- Proteins bind to lipid : Phosphatidylinositol.
  - Significance :
  - Normal gene : Prevent hemolysis.
  - mutated gene : Paroxysmal Nocturnal Hemoglobinuria.

Paroxysmal Nocturnal Hemoglobinuria

Pathology : Mutated or absent PIGA gene.

↓

During sleep (Night)

↓↓ pH

Activates complement proteins : C5-C9 (membrane attack complex)

↓

Attacks RBC membrane

↓

Hemolysis (Hemoglobinuria).

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Treatment :

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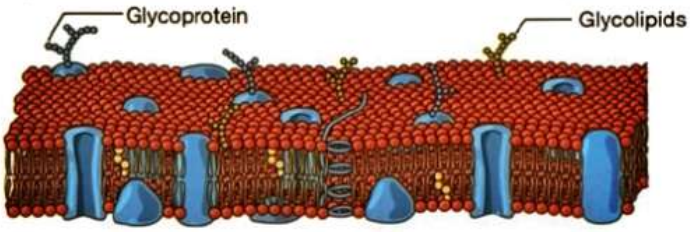
- Anti C5 monoclonal antibody : Supportive.
  - Eculizumab.
  - Ravulizumab (Long acting).
- Hematopoietic stem cell transplant : Definitive.

Carbohydrates

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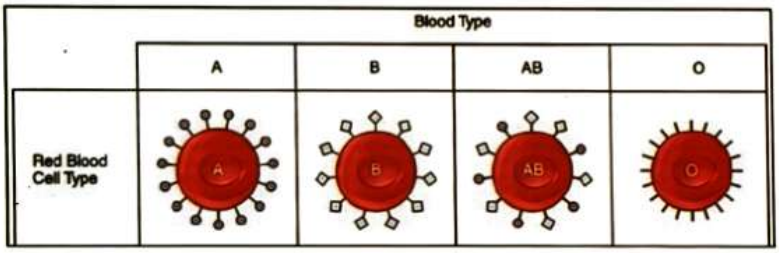
CLASSIFICATIONS :

1. Glycoprotein : Attached to proteins.
2. Glycolipids : Attached to lipids.



SIGNIFICANCE :

Glycolipids on RBC membrane : Determine blood groups.



Note :

- Membrane with highest protein : Inner mitochondrial membrane.
- In nerve cell membrane (myelin) : Lipid (80%) : Protein (20%).
- Repair of damaged cell membrane is by resealing of lipid bilayer.
- Deficiency of lecithin in pre-term baby : Hyaline membrane Disease.

Feedback